Mohammad Azmain Hossain Alfi

Computer Science and Engineering Graduate | Machine Learning, Software Engineering and Applied Research

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Personal Statement

Computer Science and Engineering graduate from BRAC University with a strong interest in artificial intelligence, machine learning, and reliable AI systems. I have experience in software development, AI evaluation, healthcare AI, and research, with projects spanning machine learning, computer vision, and AI-generated clinical documentation. I aim to build practical, trustworthy AI solutions and contribute to advancing AI innovation in Bangladesh.

Education

Bachelor of Science in Computer Science and Engineering **2021 – 2026**
BRAC University, Dhaka, Bangladesh

- Cumulative grade point average 3.04 out of 4.00, rising through the degree: 3.41 in my final year and 3.61 in my last semester, as I moved into machine learning and research.
- **Strongest results:** Machine Learning (A−), Final Year Project (A), Linear Algebra and Fourier Analysis (A), Automata and Computability (A), Digital Logic Design (A).
- **IELTS Academic 7.5** (Listening 8.5, Reading 7.0, Writing 7.0, Speaking 7.5), February 2026. Pearson Edexcel UK curriculum, taken in Bangladesh: GCE A Levels (2021) Mathematics, Physics and Chemistry, all A; IGCSE (2019) A* in all three.

Research

Two independent papers, written in 2026 and submitted in July 2026 to the IEEE Conference on Engineering Informatics (ICEI 2026, Australia track). Neither is published. Full manuscripts available on request.

Mapping the Genuine Advantage of Quantum Artificial Intelligence in Pharmaceutical Manufacturing **2026**
First author, with F. J. Afseen

- Drug companies are being urged to fund quantum computing for tasks such as inspecting tablets on a production line and screening candidate molecules. These are expensive, long-term bets, and the claim behind them had not been tested under controlled conditions.
- **Finding:** across seven public datasets, running one identical process and changing only the data, quantum models did no better than ordinary ones. I showed that two of the three standard ways of encoding data into a quantum circuit are ordinary mathematics in disguise, so they cannot win in principle, and traced the remaining gap on images to how the data was compressed rather than to the quantum model. The paper concludes that the nearer-term value of quantum computing lies in simulating molecules, not classifying data, and gives a short set of checks a company can run before funding a pilot. *Python, PennyLane, scikit-learn.*

What Changes and What Survives Across Days in Surface EMG Decoding **2026**
Co-author, with M. Kabir

- Surface electromyography reads muscle activity through electrodes on the skin and controls prosthetic hands, rehabilitation devices and wearable controllers. Accuracy falls sharply the next time the device is worn, which is one reason people stop using them. Across 63 participants in two public datasets we tested the two intuitive fixes: place electrodes on each person's strongest muscle site, and correct for electrode movement afterwards.
- **Finding: the fix is rescaling the signal, not repositioning the electrodes.** Concentrating electrodes on the strongest site raises the signal 1.7-fold within one session, but across days it loses to spreading them out, and adding more electrodes does not buy the loss back. No correction for electrode movement beat simply doing nothing. What does work is rescaling each channel from a single unlabelled calibration contraction: this recovers 44.5% of the accuracy lost between days and matches supervised retraining, without asking the user for new labels. *Python, public sEMG datasets, machine learning classifiers.*

Projects

NoteFidelity: An Open Evaluation Suite for AI-Generated Clinical Notes **2026**
Independent project, ongoing. Open source at github.com/azmainalfi22/notefidelity

- Hospitals are starting to use AI that listens to a consultation and writes the clinical note. Scoring these tools by comparing wording against a reference note means a note can score well while omitting something that matters, such as a medication dose. NoteFidelity checks whether the facts spoken in the consultation actually reached the note, and grades each missing fact by how much harm it could cause.
- I built four ways of checking a fact against a note and measured them against clinician judgement. **I set the required level of agreement before starting, did not reach it, and published that result rather than lowering the standard.** Work continues. *Python, FastAPI, Next.js, locally run language models.*

Skin Cancer Detection from Dermatology Images **2026**
University team project, team of three. Source on GitHub

- Controlled comparison of six machine learning models on 10,015 skin lesion photographs. The best image model reached 87% accuracy against 83% for models given only patient details such as age and lesion site, which measures what the photograph adds over recorded clinical data. We split the data by lesion rather than by photograph, since several images of one lesion appearing in both training and testing would have flattered the results. *Python, PyTorch.*

Chokh-e-Dekha: Civic Reporting and Follow-Up Platform **2026**
Individual project, live at chokh-e-dekha.vercel.app

- In Bangladesh, reports of damaged roads, uncollected rubbish and utility failures disappear into government offices with no way to follow up. Residents report a problem with a photo and location and track it publicly until it is resolved. It groups duplicate reports, routes each one to the department responsible, marks a problem fixed only when the original reporter confirms it, and drafts formal complaint letters in English and Bengali when an authority does not respond. *Next.js, TypeScript, Supabase (PostgreSQL).*

Pharmazine: Pharmacy Management and Point of Sale System **2026**
Started at Xactidea Ltd., later extended independently. Live at pharmazine.netlify.app

- Small pharmacies lose money by running out of a medicine and by letting stock expire, and because medicines arrive in batches with different expiry dates, tracking it by hand is difficult. Pharmazine handles sales, stock and purchasing across branches down to individual batches, reorders automatically when stock runs low or nears expiry, and separates permissions for owners, managers and staff with two-step sign-in. *React, TypeScript, FastAPI, Supabase (PostgreSQL).*
- **Robotics and embedded systems** (university, 2025): a self-balancing two-wheeled robot that reads temperature, pulse, blood oxygen and blood pressure and sends a mobile alert when a reading leaves the safe range, and a four-joint robotic arm controlled by hand movement over a wireless link, on a circuit board I designed. *C/C++, ESP32 and STM32, KiCad.*

Professional Experience

Trainee Software Developer **Oct 2025 – Jan 2026**
Xactidea Ltd., Dhaka, Bangladesh

- Built the first version of Pharmazine and delivered it to the company. The engineering problem was reliability: two staff selling at once must not be able to sell the same medicine twice, and the till has to keep working when the internet drops, which is common outside the larger cities.

Product Intern **Jun 2025 – Sep 2025**
Aspire to Innovate (a2i), ICT Division, Government of Bangladesh

- Worked on Muktopaath, Bangladesh's national online learning platform, used by more than 600,000 learners. Gathered requirements, wrote feature documentation, and tested the website and Android app, describing faults for the development team.
- Turned real support conversations into a sorted question and answer dataset for the platform's help chatbot, and ran weekly sessions on AI and cybersecurity for the team.

Technical Skills

Machine learning and data	Python, PyTorch, TensorFlow, scikit-learn, pandas, NumPy, SciPy; image classification, transfer learning, imbalanced data, natural language processing, experiment design, significance testing and model evaluation; SQL, Power BI, Tableau
Programming and web	Python, TypeScript and JavaScript, PHP, C and C++, Java, SQL; React, Next.js, FastAPI, Laravel, Node.js, REST APIs, Tailwind CSS
Databases and tools	PostgreSQL, Supabase, MySQL, Redis; Docker, Git and GitHub Actions, automated testing, Linux, Vercel, Netlify, Render; ESP32, STM32 and Arduino microcontrollers, KiCad

Certifications, Leadership and Languages

- **Artificial intelligence short courses** (Anthropic and CodePath.org, 2026): Model Context Protocol, introduction and advanced; Claude with Amazon Bedrock; Claude Code; and three on using AI tools effectively and responsibly. **Cyber Security V1**, DIKKHA Project, ICT Division, Government of Bangladesh, 2025.
- **Leadership and volunteering:** Student Ambassador for Student World Bangladesh. Elected Class Representative at BRAC University. Organised career fairs and volunteered in flood relief and blood donation drives. **Languages:** Bengali (native), English (IELTS Academic 7.5), Hindi (conversational).
- *References, full manuscripts of both papers, academic transcript and project source code available on request.*